CJC
Complete Java Classes

by Kunal Sir

Operators in Java: -

Operator are special symbols that perform specific operation on one, two or three operands and then return a result

Arithmetic Operators: We use arithmetic operators to perform arithmetic operation.

1. **+** → Addition
2. **-** → Subtraction
3. ***** → Multiplication
4. **/** → Division
5. **%** → Modulus
6. **++** → Increment
7. **--** → Decrement

Relational Operators: Relational operators in java are used to comparing two variables.

1. **x == y** → Equality operator
2. **x != y.** → In-equality operator
3. **x > y** → Greater Than
4. **x >= y** → Greater Than Equals-To
5. **x < y** → Less Than
6. **x <= y** → Less Than Equals-To

Logical Operators: Logical operators can be defined as a type of operator that helps us to combine multiple conditional statements.

1. **x && y → AND :** if both x and y are true, the result will be true.

2. **x || y → OR :** if either x or y or both are true, the result will be true.

Conditional Statements: -

Conditional statements in Java are the executable block of code (or branch to a specific code) dependent on certain conditions. These statements are also known as decision statements or selection statements in Java.

The conditional statements if, if-else, and switch allow us to choose which statement will be executed next.

Each choice or decision is based on the value of a Boolean expression (also called the condition).

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The If (if) Statement:

If we have code that we sometimes want to execute and sometimes we want to skip we can use the if statement. The form of the if statement is if (Boolean expression) statement.

If Boolean expression evaluates to true, then statement is executed.

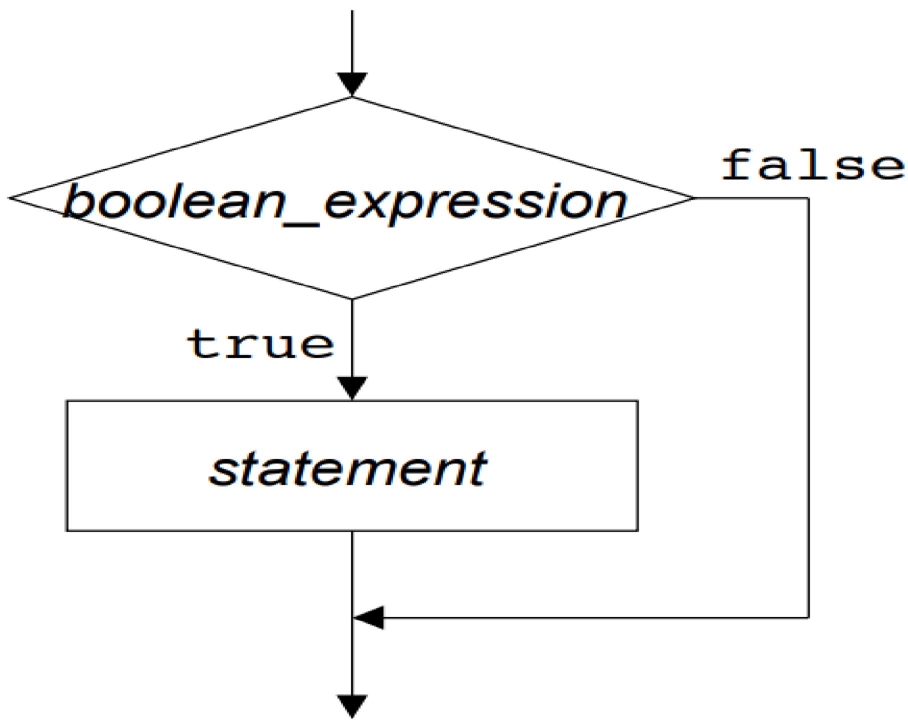
If boolean expression evaluates to false, then statement is skipped.

Note that the condition enclosed in parentheses must evaluate to true or false.

SYNTAX: -

```
if (condition) {

  // Executable code if condition is true.

}
```

The If (if) Flowchart:



The If-Else (if-else) Statement:

Use the else statement to specify a block of code to be executed if the condition is false.

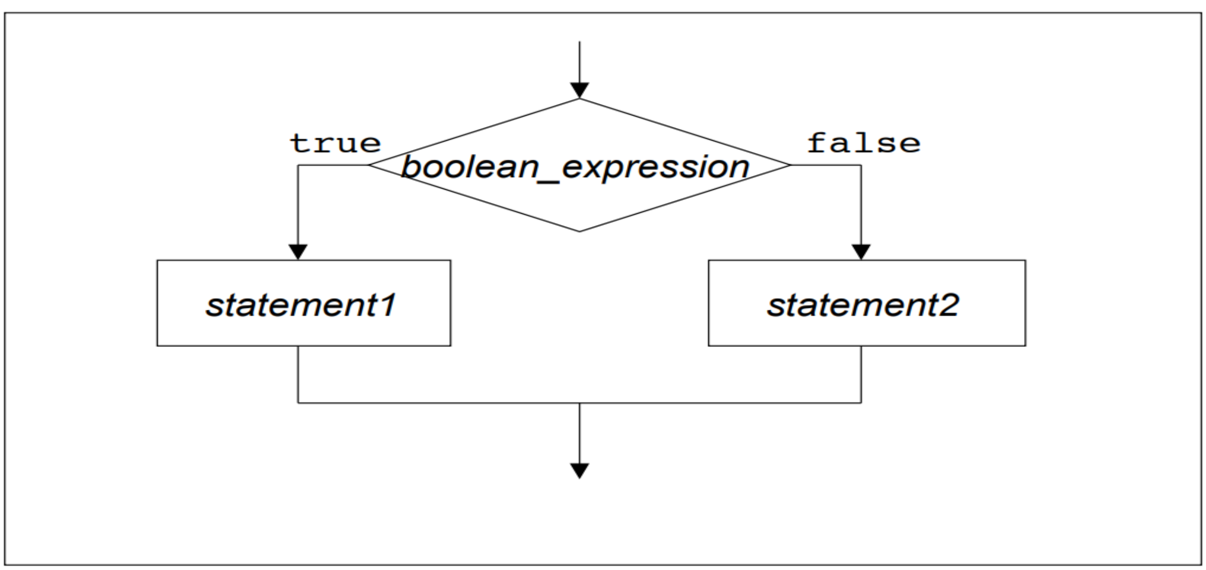
If we want to choose between two alternatives, we use the if/else statement.

SYNTAX: -

```
if (condition) {

        // Executable statement1

} else{

      // Executable statement2

        }
```

If condition evaluates to true, then statement1 is executed.
If condition evaluates to false, then statement2 is executed.

THE If-Else (if-else) Flowchart:



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The Ladder (if-else-if) Statement:

An if-else-if ladder in java can execute one block while multiple other blocks are executed.

SYNTAX: -

```
If (condition1) {
// Executable code if condition1 is true.
} else if(condition2) {
// Executable code if condition2 is true
} else if(condition3) {
// Executable code if condition3 is true
} else if(condition4) {
// Executable code if condition4 is true
} else if(condition5) {
// Executable code if condition5 is true
}

....
....

else {
 // Executable code is all the condition are false
}
```

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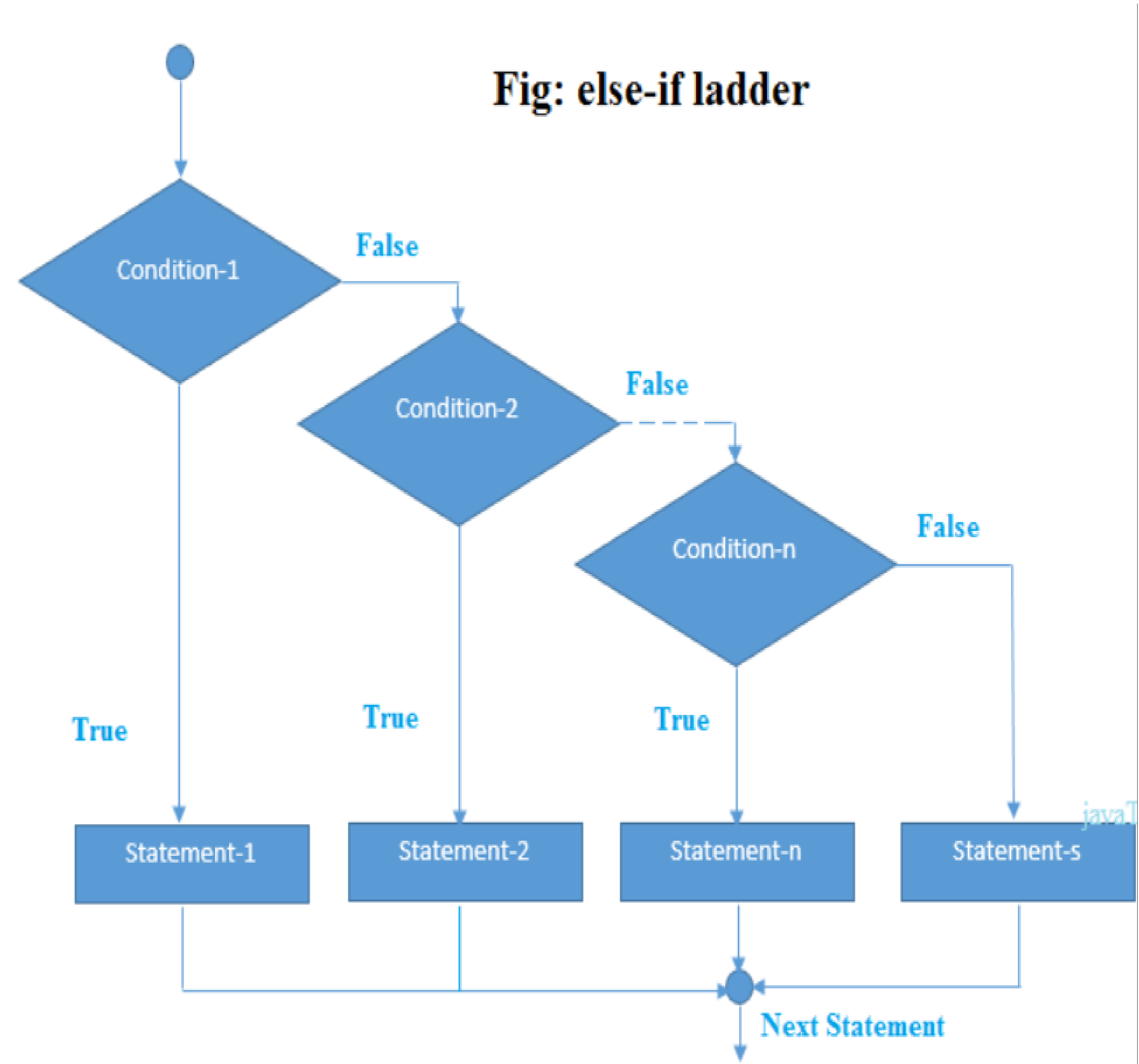
The LADDER (if-else-if) FLOWCHART:



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The Nested (if-else) Statement:

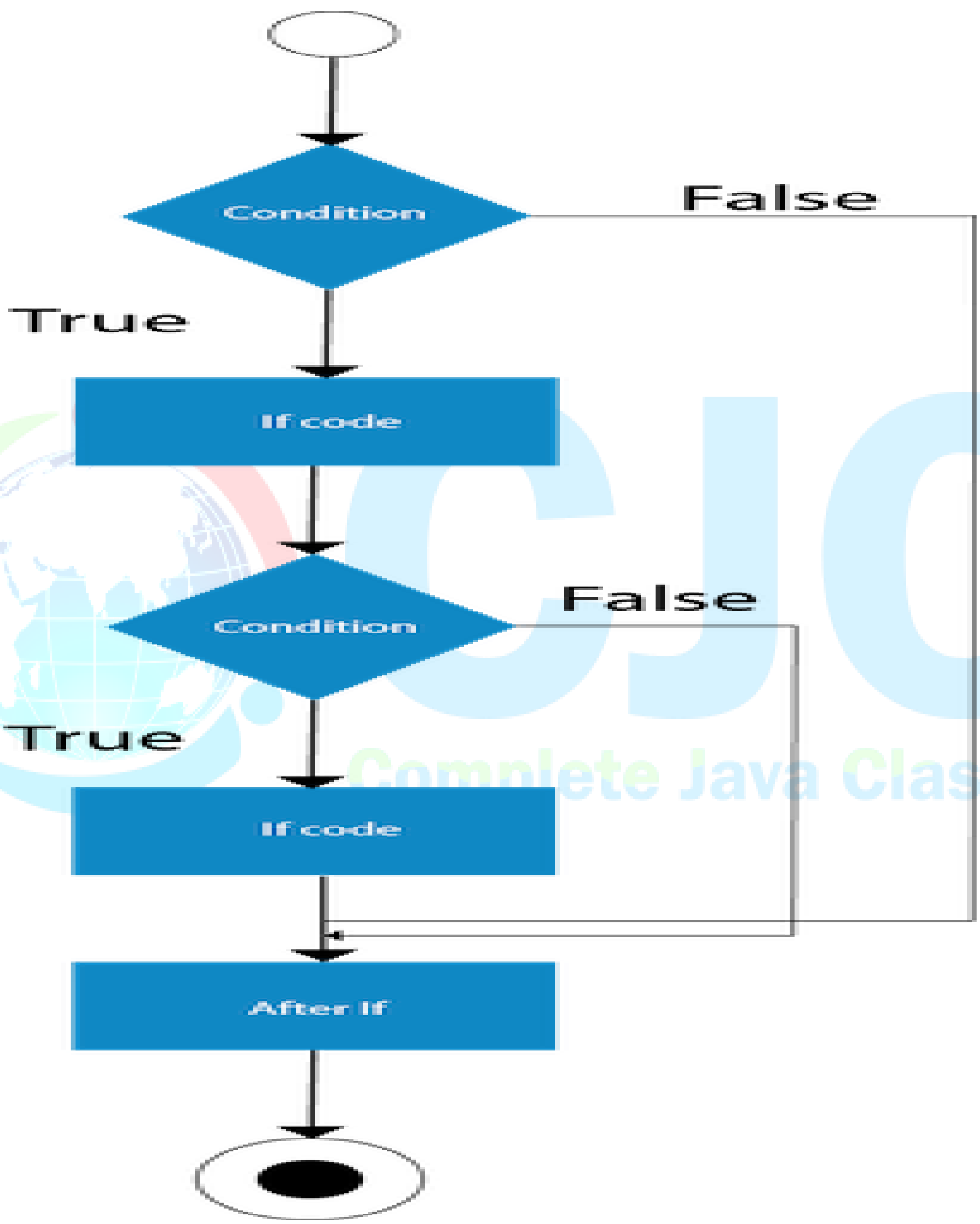
If conditions inside another if conditions are called as Nested if conditional statements in java.

In case the of nested if else conditional statements, conditions are interdependent to each other.

SYNTAX: -

```
if (condition1) {
      //Statement 1 will be executed if true
             If (condition2) {
                   // Statement 2 will be executed if true
                         If (condition3) {
                               // Statement 3 will be executed if true
                         } else {
                           // Statement 3 will not be executed if false
                            }
             } else {
                // Statement 2 will not be executed if false
             }
 } else {
 //Statement 1 will not be executed if false
}
```

The Nested (if-else) Flowchart:


Condition
False
True
If code
Condition
False
True
If code
After If